MAPPING A RESEARCH AREA WITH AI AGENTS: A WORKFLOW, AND 373 OPEN PROBLEMS IN ANALYSIS

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Abstract. We describe a workflow that lets a mathematician with graduate-level familiarity with an area enter its research literature and make research-level progress with AI coding agents, and we report what it produced in one area of analysis: Hardy, Bergman and Dirichlet-series spaces, Hankel and Helson operators, and Nehari-type theorems. The workflow has four parts: a harvested map of the area's open problems, explicit rules for directing agents, a per-problem record that carries the work from one session to the next, and review gates before anything leaves the repository. We describe each part as it was used, the failures that changed it, and the version we now recommend. In October 2026 the agents collected 373 problems, found that 64 had already been settled by others, and produced claimed answers to 141 more. Most of these are easy in hindsight. Seven are, in our judgement, of real significance, and on the hardest problems the agents produced maps of what fails rather than proofs. A blind audit of a random sample of 20 answers found no mathematical errors, but two that answer less than was asked. The work was directed by one person with a master's degree in the area who is not a research specialist. All results of ours are sketches until they appear on arXiv or in a journal, and the next stage is review by the people who asked the questions.

1. Introduction

Terence Tao has described research mathematics as climbing "a big mountain range with lots of tall mountains and lots of foothills" [55]. Human mathematicians climb one step at a time, but they can plan a route to a distant summit. In his picture, current AI tools are small jumping robots: they can sometimes "parkour their way to the top of a 6-foot wall" that a human could not climb, but they cannot plan the long route, and "the little jumping robots are nowhere near the Mount Everests of math" [55].

This paper is about the person who points the robots. Our claim concerns a method: with a workflow of the kind we describe, someone with graduate-level familiarity with an area, but without a specialist's command of any of its problems, can enter the research literature, find out what is open, and make progress that a specialist would recognise as research. We give the workflow in enough detail to be reused, the rules the author used to direct it, the ways it failed and how each failure changed it, and what it produced in one area of analysis in October 2026.

The area is the theory of Hardy and Bergman spaces on the disc and the infinite polydisc, Dirichlet series viewed as functions on the infinite torus, and the Hankel, Helson and composition operators acting on them. A typical question, and one of the results discussed below, comes from Brevig and Saksman [6]. For $0 < p < 1$ the Hardy space H^p consists of the analytic functions f on the unit disc for which $\sup_{r<1} \int_0^{2\pi} |f(re^{it})|^p \, \mathrm{d}t/2\pi$ is finite; it is not a Banach space, and many standard tools fail. Among all f in the unit ball of H^p, consider those that maximise the size of the kth Taylor coefficient. Brevig and Saksman conjectured that every such extremal function has no zeros in the disc. The conjecture is precise and checkable, it had resisted its authors' methods, and an answer would be noticed by the few dozen people who work on such problems. Most questions in our database are of this kind.

The outcome, in brief: the agents collected 373 problems, established that 64 were already solved, and produced claimed answers to 141 others. Most answers are short arguments from

Date: October 14, 2026.

known tools, results assembled from the literature, or counterexamples that are easy to check once found. Seven are, in our judgement, significant. On the problems where experts had made partial progress and stopped, the agents produced special cases, reformulations and records of what fails. The gain was largest in mapping the area and clearing its long tail, and smallest on its hardest problems.

Section 2 describes the starting point, Section 3 how the map was built, Section 4 how the work was directed, and Section 5 how it was carried from one session to the next. Section 6 lists the failures and the corrections they forced, and Section 7 follows three of them in detail. Section 8 reports what the method produced, including a blind audit, and Section 9 what it cost. Section 10 compares the work with other reports, Section 11 lists its limits, and Section 12 describes the next stage, review by the people who asked the questions. The appendices hold a sample record, an analysis of the record fields, notes on version control and files, the practices we adopted from other studies, and the agents' own reports of the main results.

2. The starting point

Three kinds of competence are easy to conflate. *Familiarity with a field* means knowing its objects, standard tools and main results well enough to read a recent paper. *Command of a problem* means knowing what has been tried on it, why each attempt failed, and which estimate would decide it; this is what a specialist brings. *Directing research* means choosing what to work on, recognising when an argument is progress and when it is churn, and judging whether a result matters. The author had the first: a master's thesis on Nehari's theorem for Dirichlet series (NTNU, 2017) and no research career in the area since. He had the third only in the weak sense of having watched it done. He had the second for no problem in the database.

The month before the campaign built the first kind of competence back up and produced its raw material. In August and September the author used the same agents to grow the thesis into a monograph, in three editions, and an online book on a teaching website. This writing did three things for the later work. It forced a shared vocabulary between author and agents, so that a brief could say "the Helson-matrix route" and mean one thing. It left a local collection of about 300 papers, which became the first source of problems. And it produced mathematics: one of the seven significant results, the critical case of a conjecture of Bondarenko, Brevig, Saksman and Seip [1], was first proved as a theorem of the monograph on 11 October, a day before the problem database existed, and only later recognised as the answer to a stated conjecture.

The agents were Claude Code (Claude Opus 5.5 as the main session, with smaller Claude models for cheap subtasks) and Codex, with OpenAI models that Codex offered under the names GPT-6 Luna, GPT-6 Astra and GPT-6.1 Sol (and earlier GPT-5.6 versions), often running at the same time in the same repository; neither tool could message the other, so the author relayed between them. The research phase ran from 3 to 14 October 2026. Its record is the version history, with about 1000 commits, and the session logs: 14 Claude Code sessions with 393 subagents, and 586 Codex sessions, 420 of them subagents.

3. Building the map

The problems came from four sources, in this order.

(1) *Stated problems in the local paper collection.* Agents searched each paper for environments labelled Conjecture, Question, Problem or Open problem, and for phrases such as "remains open" and "it would be interesting to". Each candidate was recorded with the paper, the exact location, and a quote.
(2) *The papers those cite*, with the same extraction.
(3) *Synthesis passes.* Agents read the recent papers of the core authors "between the lines". This produced *implicit* questions, which a remark suggests without posing,

and *inferences*: paper A proves one statement and paper B another, so a third should follow.

(4) *Problem lists and monographs*: workshop problem sessions, a classical problem book, and the open-problem sections of two monographs.

Table 1 follows the problems from extraction to the database. The local collection was broad, since the author's library also held books from unrelated projects, so the first catalogue contained questions from fluid dynamics, arithmetic geometry and combinatorics. Scope was then fixed by one sentence: *Hardy and Bergman spaces, Dirichlet series, analytic extremals, Hankel operators, sampling and interpolation, and directly related analytic number theory*, informally "what the people working in this area would work on". An agent applied it theme by theme and wrote down a reason for every excluded theme, for example "Euler and Navier–Stokes questions are outside the selected scope"; adjacent topics were kept only with a stated connection, such as random multiplicative functions for their link to Dirichlet-polynomial norms. Excluded records were kept in a separate report rather than deleted.

Stage (date)	Records
PDFs screened in the local collection	299
Candidate records extracted, deduplicated (10 Oct)	459
of which problem statements	399
of which incomplete candidates, dropped	60
Excluded as out of scope, with a reason per theme (11 Oct)	166
Kept in the focused catalogue, plus 2 local additions	235
first order (in the collection) / second order (cited by it)	178 / 57
Database at migration, adding 40 questions from our own book (12 Oct)	270
Added from synthesis, workshop lists and monographs (12 Oct)	72
Added during the work, mostly questions met while attacking others	31
Database	373

TABLE 1. How the database was built. Duplicates were merged at each step, so the rows do not add up exactly.

Each problem became one Markdown file with validated metadata: its theme, its kind (298 stated, 55 implicit, 20 inferences) and origin (289 asked by others, 84 by us), tags, impact and difficulty scores, a dated literature check with quoted evidence, the stage reached in our own work, and, once anyone settled the problem, how far the resolution is trusted. Section A shows a complete record. Moving from a single catalogue file to one file per problem let several agents and two different tools edit the database at once.

Difficulty as a tool for ordering work. Each problem received a difficulty score from 1 to 5. The scores were a device for deciding what to attack next and how long to persist, and we read them that way; they are not calibrated predictions. They came in two rounds. On 10 and 11 October cheap Codex agents (GPT-6 Luna), each with about ten minutes per problem, assessed 194 problems from the first catalogue: a literature check, a short attempt, and a "speculative difficulty score". Their brief warned that "a ten-minute failure to prove a statement is not evidence for a high score". On 12 October three Claude Opus 5.5 subagents scored the whole database under a common brief that defined difficulty as "effort for us with current tools" and fixed an anchor problem at each level:

Difficulty	Meaning
1	short argument from known tools; days
2	focused effort; an existing method adapts, or a computation settles it
3	needs a genuinely new idea, but a route is visible
4	hard; the natural approaches are known to fail
5	major open problem, or equivalent to one

For the 189 problems that already had a first-round score, the Opus agents kept it. Over all 373 problems the initial scores were distributed as 37, 48, 110, 145 and 33 for difficulty 1 to 5. Scores then moved as attempts failed, mostly by a mechanical rule (Section 4), so the final scores record outcomes; we use the initial scores of 12 October throughout. Even these are not blind, since they were assigned after the directed work of 3 to 11 October.

4. Directing the work

The author's part of the work was a small set of decisions, made again and again. We state them as rules, as they were eventually written into the agents' instructions.

What to attack. Problems were attacked in rounds by difficulty, from 1 upwards: "start with the low-hanging fruit". Within a round the orchestrating session picked open, unparked problems with the best impact relative to difficulty, mixing cheap problems with one or two of high impact, and skipping problems whose literature status was an unverified claim by others.

How long to persist. Each problem had a budget of working sessions (one agent attempt, typically up to an hour) equal to its difficulty. A problem at difficulty 1 to 4 that used its budget without a full answer was raised to difficulty 5, and a problem was *parked*, removed from the queue, once it reached difficulty 5 or had used six sessions without a partial result. Difficulty-5 problems were worked only in dedicated rounds with a specific new idea. Budgets first counted minutes; they were changed to sessions on the same day, because agents given 30 minutes used two to four and stopped, so time was not the binding constraint.

When to reopen. A parked problem could be reopened only for a new idea, a new paper or a new tool, and the reopening had to name it. One contractivity conjecture was unparked once, for a new reformulation, worked for a further session, and parked again by the author's decision when the reformulation reduced to the same open case.

Which model. Cheap models did literature checks and triage, and the strongest available models did attempts and refereeing. A written "research contract" of 11 October fixed this after a day in which, in the author's words, agents were "spinning": literature first with a cheap model, then a list of routes and obstructions, then an attempt, with a stronger model only for promising targets. A referee from a different model family than the author of a result was used where available.

Parallel or focused. Most problems were worked by one agent at a time, with up to about seven agents running on different problems. On a few hard problems several agents shared one problem, each assigned a different approach (*lanes*), while a coordinating session read their interim reports and redirected them, for example "three lanes have proved the same single-scale bound; stop extending it". The contrast between the two modes is one of our clearest findings: five of the seven significant results came from the week of focused work before the database existed, when the author steered one problem at a time with the asker's question in view, and none came from the multi-agent lanes.

What counts as progress. This rule changed during the work, and the change resolves an apparent tension in the method. While building the map, a reduction, a special case or a precise obstruction is progress, because the goal is to know where each problem stands. While trying to settle a problem it is not: the only progress is a theorem, or a proof complete except for one precisely stated lemma. Agents left to themselves keep producing the first kind of output after the objective has become the second. The rule we now use is to state

the current objective in every brief and to measure progress against it. "Partial results don't count" was the author's shorthand for the second regime.

When to stop and hand over. When a problem reduced to an estimate that would itself be a theorem, and further sessions only added special cases, the author stopped and wrote the reduction up for the person who had asked the question (Section 7).

5. Continuity: the record as memory

Agent sessions end often: at a context limit, a usage window, a dropped connection, or simply because the agent stops. Everything a later session needs must therefore be in the repository. The mechanism is the problem record. Its body has four parts: the *statement* as posed; a *state of play*, rewritten at the end of every session, giving what is known, the best route and the exact open subgoal; a list of *failed approaches*, one line per dead route with the reason it failed; and an append-only *attempt log*, one line per session with the route, the outcome, the reason and the next step. A session begins by reading these and the source passage and writing a three-line plan, and ends by rewriting the state of play and appending to the log. By the end, 207 records carried a list of failed approaches and the logs held about 780 entries. The rule that the reason must be precise, for example "the Schur test with weight $x^{-1/2}$ loses a log because the row sums diverge at the diagonal", did more than any other wording in the protocol.

A claimed result had to become a *solution sheet*: the problem as posed, the answer as a theorem, a complete proof with exact citations, and nothing else, built through a LaTeX pipeline that refuses files that fail a style check. A fresh agent with a fixed brief then refereed it, re-reading the question in its source, re-deriving each step, rerunning any certificate, and deciding whether the answer was full or partial.

A worked example. Ramberg's conjecture [4, 10] states that restricting a function on the bidisc to the diagonal is a contraction from the Hardy space H^p of the bidisc to the Bergman space A^p of the disc for every $p \geq 2$; it is known for even p. Its record shows four sessions by different agents over four days. The first (11 October) proved special classes and ended with "next: prove or disprove the proposed zero-free diagonal majorant lemma". The second (13 October) started from that line, searched for a counterexample at non-even p, found none, proved that extremals at even p have no zeros, and ended with "next: prove an entropy inequality, or restrict the conjecture; otherwise park". It was parked. The third session reopened it for a new idea, a Hilbert-space reformulation, proved that the vector-valued version is equivalent to a weighted transference inequality, and recorded exactly why two natural interpolation steps fail. The fourth started from that transference, proved the conjecture for all functions whose diagonal is z and, for $p \geq 4$, z^m, refuted a natural strengthening in convex order with a degree-six polynomial, and isolated the case that remains: diagonals without zeros of functions that have zeros. No session repeated an earlier route, and each began where the previous log line pointed. The conjecture is still open, and the record is now a precise account of where it stands.

6. Failure modes and corrections

The workflow above is the version we ended with. It was not the version we started with, and most of its rules exist because something went wrong. Table 2 lists the failures, what caused each, and what we changed. The playbooks published with this paper are the corrected version; this section explains the difference between the workflow as used and as recommended.

The last row deserves a word. Reading his side of the logs back, the author made the same interventions again and again: "did I ask you to stop?", "any real progress or just churn?", "partial so an expert understands it". A correction made in conversation lasted until the session ended. A correction written into the standing instructions, the agent briefs or

Failure	Cause	Change
Shallow first passes	bulk briefs; one route per problem, stop on failure	three genuinely different routes before giving up; read the log first
Cataloguing, not proving	no stated objective; obstructions look like progress	prove mode: one strategy, pushed to a theorem or to one stated lemma
Overclaiming scope	referee read the record's paraphrase, not the source	referee re-reads the source and alone sets full or partial
Duplicated parallel work	overlapping briefs	disjoint lanes and a coordinator who redirects
Unbounded numerics	background searches without limits or a target	bounded runs with progress output, deciding one named quantity
Gaming the checks	an agent told to fix lint warnings silenced the linter	humans or a second agent sample the diffs
Difficulty drifting	scores rewritten after failed attempts	initial score frozen at triage
Lost sessions	context limits, usage windows, disconnects	hand-over state at the end of every session
Sprawling files	each agent wrote its own note; drafts archived, not deleted	one notes file per problem; scratch work outside the repository
Askers not recorded	nothing in the pipeline read the field	publishing refuses an external result whose askers have not been told
Corrections not sticking	repeated in conversation only	written into the standing instructions, which agents read at the start of each session

TABLE 2. Failures of the workflow as used, their causes, and the corrections in the recommended version.

persistent memory lasted. Directing agents is largely a matter of turning repeated corrections into written rules.

Some corrections we could measure. After the referee was required to re-read the source and set the scope itself, the blind audit of Section 8.6 found two scope downgrades in 20 answers and no other errors; before it, the most common error the referees caught was an answer to a special case reported as an answer to the question. Others we could not, since the rule came too late to compare before and after on the same problems.

7. Three episodes

Outer extremals: from sweeps to a proof, and from a proof to a paper. *What happened.* On the Brevig–Saksman conjecture the agent spent its first sessions on numerical sweeps of the extremal problem. *What the author noticed.* The sweeps were accurate and did not get closer to a proof. *What changed.* He told the agent so, and asked for an argument instead. Within the session the agent reported a proof that the extremals have no zeros; two hours later a subagent extended it to every bounded linear functional on H^p, $0 < p < 1$, including singular inner factors, which gives the conjecture in a stronger form. A Codex agent estimated a Lean formalization at "months, potentially longer"; it was finished overnight. The result went to Brevig, who rewrote the proof in his own style in an afternoon at half the length, and became a coauthor. *What happened after.* Two rules: numerics must decide one named quantity, and every proof for an external question is rewritten in the asker's notation, with the idea before the computation.

The endpoint: lanes, catalogues and a hand-over. *What happened.* The endpoint $p = 4$ of Nehari's theorem in the Schatten class S_p for Hankel operators on the Paley–Wiener

space, which Bampouras singled out after proving the theorem for $p > 4$ [3], received three days of work by up to seven agents in parallel, each on one approach. *What the author noticed.* Every lane produced more special cases and more equivalent reformulations; three proved the same lemma. He called the pattern "filling $\mathbb{R}$ with $\mathbb{Q}$". *What changed.* The prove-mode brief, and disjoint mandates for the lanes. Under prove mode every route reduced to the same square-function estimate in L^1, which would be a theorem in its own right. *What happened after.* The author stopped the work and sent Bampouras a note with the reductions. The episode is the clearest instance of Tao's picture: the agents climbed many short walls around the problem and none of them led to the summit.

"Partial results don't count". *What happened.* On 13 October a campaign over difficulty-4 problems reported many problems as settled. *What the author noticed.* Read against the source, several answered an easier question: a special case reported as the conjecture, "for some p" reported as "for every p". When asked to recount, one agent wrote: "Against the original questions: 0 of these 4 fully solved. I made substantive partial progress, but selected easier subproblems instead of problems I could plausibly finish". *What changed.* A separate field for full or partial, set only by the referee after re-reading the source; partial results do not receive a resolution tier, and repeated narrowing without a new idea counts as no progress. *What happened after.* The database separates 141 full answers from 142 partial results, and the audit found two answers whose scope still had to be downgraded.

8. What the method produced

8.1. **Counts.** Table 3 gives the outcome of all 373 problems by initial difficulty. A lower score went with being settled, with an area under the ROC curve of 0.79 (FAR reports 0.69 for its difficulty scores [16]). The figure flatters the scores: they were not blind, and problems were attacked in order of difficulty, so the easy ones also got attention first. The table also shows why easy problems are easy: of the 37 rated easiest, 15 already had a published answer. Tao observed the same on the Erdős problems: those current tools can solve are "precisely the category of nominally open problems that are most likely to have been solved in the literature" [60]. The literature check also caught a problem recorded as resolved by a paper that had since been withdrawn.

Initial difficulty	Settled by others	Settled by us	Partial	Open	Total
1	15	22	0	0	37
2	12	34	2	0	48
3	20	50	32	8	110
4	10	32	91	12	145
5	7	3	17	6	33
Total	64	141	142	26	373

Table 3. Outcome of all 373 problems by initial difficulty. "Settled by us" counts claimed answers that passed one agent referee.

Figure 1 shows when the 141 answers came. The database existed only from 12 October, so everything from the directed work of 3 to 11 October is in the first bar: 30 answers, including five of the seven we rate significant. The bulk campaign of 12 October settled 45 problems, almost all of difficulty 1 to 3 and none significant. On 13 October the agents worked through harder problems: 25 of its 66 answers were rated 4 or 5, and two are significant.

8.2. **What kind of results.** We classified each of the 141 answers by *sign* (82 proofs, 42 counterexamples, 17 determinations of a value, set or order) and by how it was obtained (20 assembled from the literature, 58 routine new arguments, 44 explicit constructions or

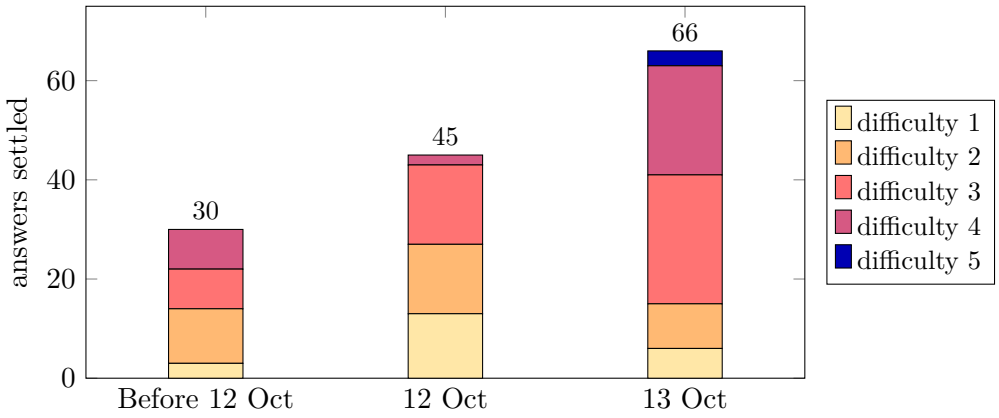


FIGURE 1. Answers settled by our agents (30, 45 and 66), by when they were recorded and by initial difficulty. The first bar collects the directed work before the database existed; its dates come from the first appearance of each write-up in the version history.

certificates, 19 substantial new arguments). Table 4 gives the cross-tabulation. Counterexamples are where the agents are strongest: 39 of the 42 are explicit constructions or certificates. Searching a space of candidates and checking each one is what agents do well, and what earlier work on machine-assisted discovery exploited [35, 47, 62]. Few proofs are substantial: 17 of 82.

	Literature	Routine	Construction	Substantial	Total
Proof	17	43	5	17	82
Counterexample	0	1	39	2	42
Determination	3	14	0	0	17

TABLE 4. The 141 claimed answers by sign and by how they were obtained.

8.3. **Significance, graded twice.** Significance is judged by the mathematics alone: *high* for an answer to a question of standing in the field, at the level of a paper; *medium* for a full answer worth a short note; *low* for a corollary, an elementary or conditional answer, or an answer to a question of little interest. The first grading was made during the work: the author chose the high results, six at the time and a seventh on 14 October, and an agent split the rest into medium and low from the records. The second was a blind re-read: five fresh agents, who had not seen the first grades, each graded about 28 records from the record and its solution sheet. Neither grading searched the literature for the standing of each question, so both measure how the results read to a model from our own records, not how a specialist would rank them.

The two gradings agree on 110 of the 141 answers (Cohen's $\kappa = 0.59$). They agree on five of the seven high results. The re-read moves two of them to medium, the counterexample to Centner's conjecture and the answer to the question of Brevig, Grepstad and Instanes, and promotes three medium results to high: an answer to a question of Guo, Ni and Zhou that also settles a 1986 question of Cole and Gamelin, a counterexample to Problem 2.3 of Saksman and Seip [56], and Suffridge's conjecture, which follows from the outer-extremals theorem and so double counts it. Elsewhere 16 answers drop from medium to low and 10 rise. No answer moves between high and low.

Neither grading looked at how each question stands in its community, which is what an expert would judge first. Four further agents therefore examined 16 boundary records: the ten graded high by either grading and six answers near the top of medium. For each record

	High	Medium	Low	Total
Proof	5	32	45	82
Counterexample	2	24	16	42
Determination	0	5	12	17
Asked by others	7	56	26	89
Asked by us	0	5	47	52
Blind re-read	8	54	79	141

TABLE 5. The 141 claimed answers by first significance grade, split by sign and by who asked the question, and the totals of the blind re-read.

they read the source passage, counted citations of the source, and searched for restatements in surveys and later papers, for attempts by others and for later answers. They spent one to three minutes per record, so this is a second opinion informed by the literature, not a referee's judgement. They graded five records high: the Bernstein–Nikolskii limits, the only grade given with high confidence, since a survey from September 2026 lists the problem first among ten open ones; the H^1 partial sums; positive Nehari; the fractional functional; and the answer to Guo, Ni and Zhou, for its bearing on the question of Cole and Gamelin. They graded the outer-extremals theorem medium, because the conjecture is recent and lightly cited, along with the answers to Brevig, Grepstad and Instanes and to Centner, and none of the six medium answers rose. Four of our seven results are graded high in all three passes. The disagreements concern the standing of a question in a small community, where citation counts are low for every paper and say little. We keep our seven and report the other grades as a measure of the uncertainty, which only the askers can remove.

8.4. **The seven significant results.** Table 6 lists them. Stated briefly: for $0 < p < 1$, every extremal of a nonzero bounded functional on H^p is outer, which proves the Brevig–Saksman conjecture in a stronger form; the extremals of the prescribed-coefficient problem of Brevig, Grepstad and Instanes have at most one zero; the sharp Bernstein–Nikolskii constants for polynomials converge to those for entire functions of exponential type, as Ganzburg and Tikhonov conjectured; the partial-sum operator on the Hardy space $\mathcal{H}^1$ of Dirichlet series has norm of order $\log N/\log\log N$; a bounded nonnegative Helson matrix need not have a bounded symbol, so the positive Nehari theorem asked about by Perfekt and Pushnitski fails; Centner's conjecture on zeros of optimal polynomial approximants fails for every $p \neq 2$; and the fractional functional $\sum a_n n^{-1/2}(\log n)^{-1/p}$ is bounded on the Hardy space $\mathcal{H}^p$ of Dirichlet series exactly when $p \leq 1$, as Bondarenko, Brevig, Saksman and Seip conjectured, with norm of order $(\log\log N)^{1-1/p}$ on polynomials of length N when $p > 1$; the case $p = 1$ was proved independently in a recent preprint of Perfekt [2]. All seven answer a conjecture, problem or question that someone else had posed formally. None is yet on arXiv.

8.5. **How the question was posed.** The database mixes conjectures, numbered problems and questions, informal remarks, and questions we derived ourselves. The records did not store this distinction, so we recovered it afterwards from each source location; the split is approximate.

How posed	Records	Others solved	We solved	Unsettled	High
Conjecture (others)	63	13	23	27	4
Problem (others)	69	18	18	33	1
Question (others)	67	12	24	31	2
Informal remark (others)	90	18	24	48	0
Our own question	84	3	52	29	0

Result	Question	Checked by	Askers told
Outer extremals, $0 < p < 1$	See [6, Conj. 1]	agent, Lean, coauthor	coauthor
At most one zero	See [5, Question 1]	agents, Lean	note prepared
Fractional functional	See [1, §3]	agent	note prepared
Bernstein–Nikolskii limits	See [8, Conj. 1.10]	agent, not filed	no
H^1 partial sums	See [56, Problem 4.5]	agent	note prepared
Positive Nehari fails	See [9, Rem. 4.11]	agents	note prepared
Approximant zeros	See [7, Conj. 7.1]	agent	no

TABLE 6. The seven significant results. "Note prepared" means the result is in a note for Brevig, an asker or a coauthor of the askers in each such case, which has not yet been sent.

Our own questions were settled most often, 52 of 84, and none of those answers is significant. Informally posed questions were the least often settled and produced nothing significant. We now record the form of each question when it is harvested.

8.6. **A blind audit of a random sample.** To estimate how far the counts can be trusted, we drew a random sample of 20 of the 141 claimed answers and 10 of the 64 problems marked as settled by others. Five fresh referee agents (Claude Opus 5.5) each took a group. They saw only the question as posed and the solution sheet or the cited paper, never the working notes or earlier verdicts, and spent 15 to 20 minutes on each.

Of the 20 answers, 18 were judged correct and complete, 2 correct but answering less than was asked, and none was flawed; with 20 answers this bounds the rate of flawed answers by about 17% at 95% confidence. One downgrade answered one part of a two-part question; the other is unconditional only for part of the parameter range and relies for the rest on an unrefereed preprint, as it discloses. The audit found more problems in the bookkeeping than in the mathematics: in five of the 20 records the metadata did not match the sheet, for example an unrefereed dependency left out of the field meant to record it. Of the 10 problems marked as settled by others, 8 were confirmed and 2 are only partly settled by the cited work.

The referees come from the same model family that produced many of the answers, 15 to 20 minutes suffices to re-derive a short argument but not to referee a long one, and the sample is small. The audit supports the claim that the arguments the workflow produces are usually sound. It does not replace review by people in the field.

8.7. **The unsettled problems as a map.** Of the 373 problems, 168 remain unsettled: 142 with partial results of ours and 26 without. They differ from the settled ones in kind.

	Unsettled (168)	Settled (141)
Initial difficulty 4 or 5	126	35
Asked by others	83%	63%
Impact 4 or more	19%	4%
Partial progress by others already in the literature	50%	21%
Inferences of ours	1	16

The unsettled problems were asked by others, they matter more, half had already been attacked in the literature, and they ask for a characterisation, an exact constant or a uniform estimate rather than a single construction. Their main obstructions, in our reading of each state of play, were that only special cases were proved (59 records), that the problem needs an estimate that is itself a known hard problem (37 records), that the question is open-ended or needs reformulation (23), and that every method loses a logarithm or a power at an endpoint (22). About 25 records reduce to three underlying problems: an L^1 endpoint estimate behind

Schatten-class Nehari theorems, the value of one sharp constant in a multiplicative Hilbert-type inequality, and the unknown extremal functions of point evaluation in Paley–Wiener spaces. Settling any one of these would move many records at once.

For someone entering one of these problems, the record is probably worth more than our proofs: a state of play, a list of the routes tried and why each failed, and the partial results, which include 90 special cases or subranges, 13 correct orders of growth without the constant, and 12 reductions to equivalent reformulations. Mathematics has almost no venue for such records, and Tao names "the publication of negative or partial results" among the infrastructure the field now needs [57]. We would release the record for any problem, labelled as a sketch, to anyone working on it.

9. Costs and requirements

Our cost data are rough. For the billing month from 8 September to 8 October, Codex use came to about $1,200 and Claude Code to about $300, both in API-equivalent terms; the month includes the writing of the monograph and other projects, and covers only the first days of the research phase, so these figures indicate an order of magnitude, not the cost of the campaign. Most of the Codex figure was cache reads by long orchestrating sessions that re-read large contexts. Cheap models for literature and triage, and strong models for attempts and refereeing, was the right split.

The other requirements were these. A person with graduate-level familiarity with the area, to read every result before it went out, to recognise when a result mattered, and to contact the askers. A local paper collection, with library access for the papers not on arXiv. A repository with version control, a lint-gated LaTeX build and a validated record format; we publish ours with this paper. And time: the research phase took twelve days, after about six weeks of writing that built the collection and the vocabulary. The scarcest resource was not compute but the author's attention, which went into reading results, redirecting agents and writing to experts.

10. Related work

Several groups have run AI systems over whole lists of open problems in the past year (Table 7). The lists differ widely in difficulty, the meanings of "resolved" differ, and so does how far the results were verified.

Study	Problems	Directed by	Outcome	Verification
Gemini on Erdős problems [17]	700 open	expert team	13 addressed, 4 novel	AI filter, experts
Formal proof search [27]	353 Erdős; 492 OEIS	research team	9; 44 resolved	Lean
OEIS Open [11]	492 OEIS	one researcher	147 resolved	Lean
Provers and verifiers [23]	66 conjectures	research team	about a third	verifier agents
QED [25]	18 expert-chosen	experts	5 original works	verifier, experts
FAR pipeline [16]	4717 harvested	research team	598 candidates	sent to authors
UQ [28]	500 questions	research team	15% pass validation	validators, community
This report	373 harvested	one non-specialist	64 known; 141 claimed; 7 significant	agent referees; 2 in Lean

Table 7. AI systems run on lists of open problems: the list, who directed the work, the reported outcome, and how it was verified.

The closest study to ours is the Gemini case study on the Erdős problems, which reached the same conclusion: "the 'Open' status of the problems was through obscurity rather than difficulty" [17]. Of the 13 problems it addressed, 9 turned out to be solved already in the literature; of 200 candidate answers it graded, 68.5% were flawed and only 6.5% meaningfully correct, because most technically correct answers had misread the question. The FAR pipeline [16] is the nearest match to our method, at about fifteen times our scale: it extracted 6453 candidate problems from 5245 combinatorics papers and sent 77 potential resolutions to their authors. Like us, its authors identify the selection of problems and the review of the results as the bottlenecks. The authors of the OEIS study note that their conjectures "are of uncertain mathematical significance, and most have likely received little previous attention" [11]. An early test with five easy conjectures found the same pattern as ours: the easy ones fell, and the one that needed two papers combined did not [19]. Individual resolutions from such sweeps have been written up with care [13, 26], and one Erdős prize problem thought open for half a century turned out to have been settled three decades before it was posed [12].

Problem lists are now also built automatically. DeepMind's formal conjectures collect 2615 statements in Lean, including 1029 open conjectures [18]. OpenConjecture extracts conjectures from new arXiv papers and ranks them by apparent tractability [34]. FrontierMath's open-problems pilot sets out to provide a "denominator" for AI contributions [32], and MathArena tracks performance on questions from recent papers [24]. The Erdős problems database [42] and its AI wiki [43] are the closest analogue to our database; the wiki warns that "absence of past progress may reflect obscurity rather than difficulty".

Gowers writes that "if a problem can be solved by means of a relatively standard argument, it is highly likely that an LLM will be able to find and use that argument", while its suggestions on harder problems often "sound promising until I think about them carefully" [33]. Case studies from industry [15, 30] and from individual groups, including one in functional analysis [14] and one with a consumer model [29], report the same division of labour: models search for proofs, and experts close them. Earlier machine-assisted discovery found constructions and counterexamples by search [35, 41, 47, 48, 62], and experiments with frontier models as research assistants are collected in [39]. Formal proof [36, 37, 51, 53, 54] and evaluation [40, 44–46, 50] form the background. Bloom reports "a wave of AI-produced solutions provided with no explanation" [31], and Kallel and El Louadi separate the validity of a derivation from whether the statement means the intended question and whether the result matters; "only the first is mechanizable" [21]. A survey of arXiv mathematics finds substantive AI use concentrated in combinatorics rather than analysis [20], and long agent runs drift in the ways we saw [22]. On the goals of mathematics beyond problem solving we follow Thurston [61], Gowers [49], Tao [57, 59] and Bessis [38].

Three differences matter for reading our numbers. Our problems were harvested, not curated: assembled by agents from papers, problem sessions and monographs, including informal remarks and our own derived questions, which gives a broader and messier denominator than a fixed public list. The area resists formalization: Hardy spaces, Schatten classes and Dirichlet series in infinitely many variables are far from current formal libraries, so apart from two Lean formalizations our results rest on agent referees, certificates in exact arithmetic, and the people who asked the questions. And the director was not a specialist. The OEIS study was also run by one researcher; what differs here is that the author had no command of any problem in the database before the work, which is closer to how a graduate student or a mathematician from a neighbouring field would use these tools.

11. Limits

This is one person's experience in one area, over twelve days of research after six weeks of preparation, without a control group. We list the limits we are aware of.

Self-grading. Difficulty, kind and significance were assigned by the agents and by the author. The blind re-read used the same model family and did not consult the literature.

Same-family review. Most answers were refereed by the model family that produced them, and the audit used it too.

Contamination. An agent may have seen a solution in its training data, or found it in a paper our literature check missed. The check is bounded and dated, and it found 64 problems already solved, which suggests that it misses some.

Selection. The problems were harvested from a collection that was built for writing a monograph, problems were attacked in order of difficulty, and the initial difficulty scores were assigned after a week of work on some of them.

Measurement. The data on the first week are coarse, because the version history became fine-grained only on 12 October (Section C). Cost figures cover a billing month, not the campaign.

Transience. The models will be superseded within months, and the area has a dense literature of recent preprints, which made the literature check unusually productive.

12. Next stage: expert review

Producing a candidate proof was cheap; checking it was the bottleneck. This matches Tao's analysis of the problem-solving pipeline, in which generation is followed by verification, exposition, acceptance and absorption into the standard theory, and in which AI threatens "proof indigestion" [57]. Agent referees caught real errors: a bound used outside its range, a missing hypothesis, and above all overstated scope. They rarely asked whether the claim was the right one.

Every result of ours is therefore labelled a *sketch* until it appears on arXiv or in a journal, however it was checked. Before anything is posted, the people who asked the question are told, following the Leiden Declaration on attribution, disclosure and the humanity of authorship [52]. The questions belong to those who asked them. The proofs and computations were produced by AI agents and checked by the author, and following Tao's rule of thumb [57], no result should be published until a human can give a clear, expert-level talk on it. Proofs for external questions are rewritten in the asker's notation, with the idea before the computation, and sent to the asker; several groups have been contacted.

The next stage of this work is that review. The people who asked the questions will decide which answers are correct, which are new, and which matter, and this paper will be updated as they respond. The significance grades above are our best estimate until then, and we expect them to move.

Tool and computational resource disclosure

The work used Claude Code from Anthropic, with Claude Opus 5.5, Sonnet and Haiku, and Codex from OpenAI, with several GPT models; Python with mpmath, SymPy and cvxpy for numerics and certificates, Lean for two formalizations, and LaTeX built through a lint gate. This paper was drafted with Claude Code from the project's records and logs and revised by the author. The agent instructions, templates, tools and the anonymised data behind every count are available in the accompanying process repository.

Appendix A. A problem record

Each problem is one Markdown file whose YAML header holds the data and whose body holds the memory between sessions. This is the record of a question of Bayart and Yao, abridged; long fields are cut with [...].

```
id: BY-8.7
title: "Minimal commutant for phi = lambda z^p psi (p>=2) on H^2(D):
  can the zero-free assumption on psi in Corollary 8.5 be dropped?"
kind: stated                      # stated | implicit | inference
```

```
origin: external                  # external | ours
theme: composition
source:
  paper: "arXiv:2406.19737"
  location: "Question 8.7, p. 33"
  posed_by: [Bayart, Yao]
literature:
  status: open
  checked: 2026-10-13
  evidence: ["arXiv:2406.19737v1 (only version) [...]"]
ours:
  stage: public
  result_scope: full
  sign: proof
  artifacts: [research/composition/solutions/BY-8.7.tex]
  review: "independent agent referee 2026-10-13: correct [...]"
  public: papers/bayart-yao-questions.pdf
  told_authors: null
resolution: {tier: ours-public, ref: "BY-8.7.tex, Theorem 1",
             checked: true, conditional_on: []}
impact: 2
difficulty: 2
scores_note: "[...] difficulty 3->2 after split from BY-8.11 [...]:
  short argument (local Boettcher coordinate [...])"
tags: [setting:disc, space:hardy, object:composition,
       type:existence, method:construction]
effort: {sessions: 1, last: 2026-10-13, parked: false}
```

The body has the four sections described in Section 5. At harvest an agent filled the identity and the source: `id`, `title`, `kind`, `origin`, `theme` and `source`. At triage a cheap agent filled `literature` and the scores. During attacks the working agent updated `ours.stage`, `ours.artifacts`, `ours.sign`, the `effort` counters and the body. A referee set the review, the scope (`ours.result_scope`) and the resolution. Only the author set `ours.public` and `ours.told_authors`. Validation by `make problems` rejected unknown fields, invalid values and inconsistent combinations, such as a resolution tier on a partial result.

Appendix B. How the record fields were used

Most fields were added one at a time over 12 October, each answering a request from the author: a difficulty score “to give us some starting points”, “different levels of solved” (the resolution tiers), a way to say which papers bear on a problem, and, a day later, whether we or someone else had asked the question. Over all 772 commits that touched the database we counted how many changed each field.

Field	Commits changing it
State of play (prose)	615
Attempt log (prose)	566
Stage	532
Session count and date	about 400
Failed approaches (prose)	322
Related papers	319
Artifacts	304
Literature evidence	275
Review	273
Full or partial	230
Origin, impact, tags	1–12
Authors told	3
Claim basis, abandonment reason, aliases	4 each

An edit count measures how often a field changed, not whether it mattered. The origin of a question was set once, in one commit, and it turned out to be one of the most informative fields in the analysis; a field set once can be essential. What the counts do show is which fields were part of the daily work: the prose memory between sessions and the few fields that move a problem through the pipeline. Two observations were clearer. The free-text justification of the scores was rewritten in 426 commits, because escalations were appended to it, and so it went stale. And the field recording that the askers had been told was changed in only three commits, because nothing read it. The lesson we draw is that every field needs a consumer, a step of the pipeline or a decision that reads it, and a field without one is not kept up to date. The schema we now recommend freezes the score justification at triage, logs escalations in the attempt log, adds the form of the question, a verbatim quote, the reviewing model and a frozen initial difficulty, and makes the publishing step refuse a result whose askers have not been told.

Appendix C. Version control and files

Version control was the research log, but only once we used it. The repository was managed with Jujutsu (`jj`), which recorded 2409 operations and 1007 commits. Of the operations, 1944 fall on 12 and 13 October, after the rule that each problem record is committed together with the work it describes. The directed work of 3 to 11 October, which produced five of the seven significant results, left only 136 operations; those results can be dated only by when their write-ups first appeared. Everything quantitative in this report comes from the fine-grained part of the history. Jujutsu's automatic snapshots meant that no edit by an agent was ever lost, and cheap small commits kept changes per record. Our playbooks therefore default to Jujutsu with a git backend and require a commit per problem from the first day.

Failed attempts sprawled. The rule that TeX files hold results only, with failures recorded in one line of the problem record, was adopted but not enforced. The research folders ended with about 350 Markdown notes, 575 TeX files, 260 scripts and 140 data files; 107 records list five or more artifacts, and one hard problem accumulated 31 files. Later agents had to read through the pile to avoid repeating routes, and sometimes did not. The playbooks now require one notes file per problem, scratch work outside the repository, scripts in the repository only when a claim depends on them, and deletion of superseded drafts, which the history keeps.

Appendix D. What the literature changed in our process

We read the closest studies in full and changed our playbooks where they had learned something we had not.

Read the question before answering it. Copy the source's conventions into the record, check a restatement against known special cases, and treat an easy answer as a sign of misreading: most technically correct answers in the Gemini study were vacuous [17], and AlphaProof's agents proved variants the askers had not meant [27].

Blind, cross-family referees. The referee first sees only the question and the proof, as in QED, whose verifier accepted 17 of 214 candidate proofs and had all 17 confirmed by experts [25]; same-family validators overrate their own answers [28]. Review is capped at two rounds, after which the problem goes to the human, since revisions converge on errors the reviewer can no longer see [15].

A word-by-word statement check, with a list of the ways a proved statement drifts from the posed one [21, 25], and a citation audit that quotes every cited result.

Outcome labels: known, new, fixed statement, or nothing [16]; claimed answers split into flawed, technically correct and meaningfully correct [17]; asymptotic results without explicit thresholds counted as partial [23].

Neutral briefs: "prove or disprove", since an agent told to prove a false statement papers over the gap [30]; and repeated narrowing to a smaller question without a new idea counts as no progress [33].

Human sign-off before release: the human confirms that the statement matches the question [21], writes or approves an exposition they understand [31], could give a talk on it [57], and adds a disclosure section [52].

Appendix E. The results in the agents' words

The session logs record what each agent wrote when it reported a result. We quote them verbatim, with model and time (UTC); trims are marked [...] and our insertions are in square brackets. The fractional functional is not included: it was proved as a theorem of the monograph while it was being written, and recognised as the answer to a conjecture only later.

Result	First reported	Found by	Sign
Outer extremals	8 Oct, 13:56	Claude Opus 5.5	proof
At most one zero	8 Oct, 20:49	GPT-6 Luna (Codex)	proof
Approximant zeros	10 Oct, 21:46	GPT-6.1 Sol; Claude	counterex.
Bernstein–Nikolskii limits	11 Oct, 19:13	Claude Opus 5.5	proof
Fractional functional	11 Oct, 19:27	monograph (Codex)	proof
H^1 partial sums	12 Oct, 22:21	GPT-6.1 Sol (Codex)	proof
Positive Nehari	12 Oct, 22:44	GPT-6 Astra	counterex.

Table 8. When the seven significant results were first reported. Each was later checked by at least one other agent; the first two are also Lean-verified.

Outer extremals.

> I believe I've proved [the nonvanishing conjecture of Brevig and Saksman]: for $0 < p < 1$, every extremal for $C(k, p)$ has no zeros in the disc. [...] Only I have checked the proof, plus numerics at $k = 2$; no one else has. —Claude Opus 5.5, 8 October, 13:56

> Done—the full theorem is now Lean-verified: every extremal for a nonzero bounded complex-linear functional on H^p, for every real $0 < p < 1$, is outer. —GPT-6.1 Sol, 10 October, 00:54

At most one zero. A Codex subagent found the key step; the deciding idea was to fix one perturbation phase rather than average over phases:

> The full real-direction Hessian yields a direct contradiction for any extremal with at least two zeros. [...] I checked the projections, the cancellation of the [...] terms, and admissibility of the perturbation. I found no gap in the argument. —GPT-6 Luna, 8 October, 20:49

The idea was relayed to the Claude session within minutes, which then reported that it had "reached the same conclusion independently". We cannot verify that claim, and we credit the Codex subagent.

Zeros of optimal polynomial approximants. At 21:58 the author asked how close the agents were to the full result:

> We're not close to the full result if that means interior zeros for every $p \neq 2$. What we have is solid for p close to 1, and nothing yet for p near 2 or above it. —Claude Opus 5.5, 10 October, 21:58

Twenty-seven minutes later a subagent reported "there is no obstruction", and the main session wrote:

> Outer targets now work for $p > 2$ as well. Agent A found the construction, and I checked it independently. So for every $p \neq 2$ there is an outer target (even a polynomial one) whose degree-one OPA has a zero in $\mathbb{D}$. —Claude Opus 5.5, 10 October, 22:25

Bernstein–Nikolskii limits.

> [The finite-q problem] is proved: for every $s \geq 0$ and $0 < p < q < \infty$, $\lim A^{(s)}_{p,q,n} = D^{(s)}_{p,q}$, and the real version holds too. I checked the proof step by step; it has not been refereed. —Claude Opus 5.5, 11 October, 19:13

A second subagent then checked priority and found the problem still listed as open in a survey from two weeks earlier.

H^1 partial sums.

> The draft now proves a lower bound of order $\log N / \log\log N$ for every sufficiently large N, matching the published upper bound. The key step is turning product entropy into the missing L^1 output bound [...] subject to independent review. —GPT-6.1 Sol, 12 October, 22:21

Positive Nehari. The same agent declined to count a numerical candidate before it had a certificate, and four minutes later had one:

> I'm not counting the new [...] numerical candidate as a breakthrough—it still needs a rigorous certificate. —GPT-6 Astra, 12 October, 22:40

> [The record] now has a rigorous counterexample certificate. For an explicit measure on two points in the real three-dimensional cube, the certified scalar norm exceeds the full Hilbert embedding norm. [...] This is a proved candidate awaiting independent review, rather than an 'almost proved' lead. —GPT-6 Astra, 12 October, 22:44

The agents on their own work.

> I did a bounded first pass, not an all-out attempt. [...] I tested one route per open problem, often a relatively elementary construction, and usually stopped when it failed. [...] 14 failed routes do not demonstrate that I seriously exhausted the mathematical possibilities. —GPT-6 Astra, 13 October

> When everything you try collapses onto the same estimate, it usually means the problem needs a new idea, not more effort. —Claude Opus 5.5, 13 October

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